CS F364: Design & Analysis of Algorithms

Quiz 5 — June 12, 2026

Coverage: Lectures 1–14

> ℹ Note
>
> **Instructions:**
>
> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question: LCS — Dynamic Programming

Let $c[i, j]$ be the length of the LCS of prefixes $X[1..i]$ and $Y[1..j]$.

For strings $X = \texttt{ABCD}$ and $Y = \texttt{AXCD}$:

(a) [2 marks] Write the complete recurrence relation for $c[i, j]$ for all $i, j > 0$. Clearly state the base case and both cases ($x_i = y_j$ and $x_i \neq y_j$).

(b) [2 marks] Fill in the DP table below. The first row ($i = 0$) and first column ($j = 0$) are already given.

(c) [1 mark] What is the length of the LCS? What is the actual LCS sequence?

Solution

Part (a) — Recurrence Relation

Base case: For $i = 0$ or $j = 0$, $c[i, j] = 0$.

Recurrence: For $i, j > 0$:

$$c[i,j] = \begin{cases} c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i-1, j],\ c[i, j-1]) & \text{if } x_i \neq y_j \end{cases}$$

$c[i,j]$	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$

Part (b) — Completed DP Table

$c[i,j]$	$j=0$	$j=1$	$j=2$	$j=3$	$j=4$
	$\emptyset$	**A**	**X**	**C**	**D**
$i=0$ $\emptyset$	0	0	0	0	0
$i=1$ **A**	0	**1**	**1**	**1**	**1**
$i=2$ **B**	0	**1**	**1**	**1**	**1**
$i=3$ **C**	0	**1**	**1**	**2**	**2**
$i=4$ **D**	0	**1**	**1**	**2**	**3**

Computation walkthrough:

- **Row 1 (A):** $x_1 = A$, compare with y_j:
  - $j=1$ (A): $A = A \Rightarrow c[1,1] = c[0,0] + 1 = 1$
  - $j=2$ (X): $A \neq X \Rightarrow c[1,2] = \max(c[0,2], c[1,1]) = \max(0,1) = 1$
  - $j=3$ (C): $A \neq C \Rightarrow c[1,3] = \max(c[0,3], c[1,2]) = \max(0,1) = 1$
  - $j=4$ (D): $A \neq D \Rightarrow c[1,4] = \max(c[0,4], c[1,3]) = \max(0,1) = 1$
- **Row 2 (B):** $x_2 = B$, all y_j are A, X, C, D — none match B:
  - $c[2,j] = \max(c[1,j], c[2,j-1])$ — all cells $= 1$
- **Row 3 (C):** $x_3 = C$, matches $y_3 = C$:
  - $j=1$ (A): $C \neq A \Rightarrow c[3,1] = \max(c[2,1], c[3,0]) = \max(1,0) = 1$
  - $j=2$ (X): $C \neq X \Rightarrow c[3,2] = \max(c[2,2], c[3,1]) = \max(1,1) = 1$
  - $j=3$ (C): $C = C \Rightarrow c[3,3] = c[2,2] + 1 = 1 + 1 = 2$
  - $j=4$ (D): $C \neq D \Rightarrow c[3,4] = \max(c[2,4], c[3,3]) = \max(1,2) = 2$
- **Row 4 (D):** $x_4 = D$, matches $y_4 = D$:
  - $j=1$ (A): $D \neq A \Rightarrow c[4,1] = \max(c[3,1], c[4,0]) = \max(1,0) = 1$
  - $j=2$ (X): $D \neq X \Rightarrow c[4,2] = \max(c[3,2], c[4,1]) = \max(1,1) = 1$
  - $j=3$ (C): $D \neq C \Rightarrow c[4,3] = \max(c[3,3], c[4,2]) = \max(2,1) = 2$
  - $j=4$ (D): $D = D \Rightarrow c[4,4] = c[3,3] + 1 = 2 + 1 = 3$

Part (c) — LCS

LCS length: $c[4,4] = 3$

LCS sequence: `ACD`

(Traceback: $c[4,4]$ came from $c[3,3]+1$ where $D = D$; $c[3,3]$ came from $c[2,2]+1$ where $C = C$; $c[2,2]$ came from $c[1,1]$ (max); $c[1,1]$ came from $c[0,0]+1$ where $A = A$. Reading the matched characters in reverse: D, C, A $\rightarrow$ "ACD".)